

Email Security Protocols

Module 19

Duration: 31:18

Video: <https://www.youtube.com/watch?v=0PUVjKXOiV8>

Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Prof Yogesh Chaba, Department of Computer Engineering and IT, Guru Jambheshwar University of Science & Technology, Hisar

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0:00 [Music]

0:19 Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar

0:27 University of Science and Technology Hisar harana in this lecture we will

0:34 study about email security protocols first of all please see the contents

0:41 which I shall cover after covering brief introduction about email security I

0:48 shall take up different email security protocols like pretty good privacy pgp

0:56 privacy enhanced mail pem and secure

1:02 multi-purpose internet mail extension s mine in last we will compare all these

1:12 protocols electronic mail most commonly referred to as email since

1:19 1993 is a method of exchanging digital messages from one sender to one or more

1:28 recipients email operates across the internet or across other computer

1:38 networks email privacy is the broad topic dealing with issues of

1:45 unauthorized assess and inspection of electronic mail this unauthorized assess

1:54 can happen while an email is in transit as well as when it is stored on email

2:02 server or on a user computer email has to go through

2:09 potentially untrusted intermediate servers before reaching its destination

2:17 and there is no way to tell if it was assessed by an unauthorized user or

2:26 not email system is different from a letter sealed in an envelope in which by

2:34 close inspection of the envelope it might be possible to tell if someone

2:40 opened it or not in that sense an email is much like a postcard whose contents

2:49 are visible to everyone who handles it there are certain technical methods that

2:57 make unauthorized assess to email difficult if not

3:04 impossible however since email messages frequently cross Nation boundaries and

3:11 different countries have different rules and regulations so email privacy is a

3:20 complicated issue this problem has stimulated several people and groups to apply the

3:27 cryptographic principles to produce secure email now we will study few
3:36 common widely used email security systems pretty good privacy
3:44 pgp privacy enhanced mail pem and secure
3:51 multi-purpose internet mail extensions s mine first of all let us see pretty good
4:00 privacy pgp short for pretty good privacy is a public key encryption
4:08 program originally written by Phil Zimmerman in
4:15 1991 over the past few years pgp has got thousands of adherent supporters all
4:23 over the globe and has become a def facto standard for encryption of email
4:30 on the internet released in 1991 pgp is a complete email Security
4:40 package that provides privacy authentication digital
4:46 signatures and compression all in an easy to use
4:53 form furthermore the complete package including all the source code is
5:00 distributed free of charge via the Internet due to its quality zero price
5:09 and easily availability on Unix Linux Windows and Mac operating system
5:17 platforms it is widely used today pgp can be used to send messages
5:27 confidentially for this purpose pgp combines symmetric key encryption and
5:34 public key encryption the message is encrypted using a symmetric encryption
5:42 algorithm which requires a symmetric key each symetric key is used only
5:51 once and is also called a session key the message and its session key are sent
6:00 to the receiver so that they know how to decrypt the message but to protect it
6:07 during transmission it is encrypted with the receiver's public key only the private
6:16 key belonging to the receiver can decrypt the session key pgp supports
6:24 message authentication and integrity checking the letter is used to detect
6:31 whether a message has been altered since it was completed and the former to
6:38 determine whether it was actually sent by the person or entity claimed to be the
6:46 sender because the content is encrypted any changes in the message will result
6:54 in failure of the decryption with the appropriate key
7:01 7 minutes, 1 second the sender uses pgp to create a digital signature for the
7:08 message with either
7:15 the RSA or DSA algorithm to do so pgp
7:23 computes a hash also known as a message digest from the plain text and then
7:31 creates the digital signature from that hash using the senders private
7:41 key to see how pgp works let us consider the example shown in figure here Alys on
7:48 the left hand side wants to send a signed plan text message P to Bob on
7:58 right hand side of figure in a secure way both Alys and Bob have RSA type of
8:07 private and public key d and e let us assume that each one knows others public
key Alis starts out by sending message P by invoking the pgp program on her

8:16 computer by first hashing her message P using md5 and then encrypts the
8:25 resulting hash using her private RSA key B A this signed hash of p is
8:34 concatenated with original message P resulting P1 which is transmitted after
8:42 compressing by zip program giving output p1.
8:49 Z on the receiver side Bob receives p1.
8:54 Z message which is given to international data and encryption algorithm idea
module for
9:04 decryption when Bob eventually gets the message he can decrypt the hash with
9:11 Ellis public key e and verify that the hash is correct after decompressing it
9:19 Bob separates the plain text from the encrypted hash and decrypts the hash
9:25 using Alis public key if the plain text hash agrees with his
9:33 own md5 computation he knows that P is the correct message and it came from
9:41 ellys after verification of hash function Bob reverses the base 64
9:48 encoding and decrypts the idea key using his private RSA
9:56 key it is worth to note that RSA is only used in two places here to decrypt the
10:06 128bit md5 hash and to encrypt the
10:11 128bit idea key although RSA is slow it
10:17 has to encrypt only 256 bits not a large volume of data thus pgp provides
10:27 security compression and a digital signature and does so in a much more efficient
10:36 way pgp supports four RSA key lens it is up to the user to select the one that is
10:46 most appropriate the length are number one casual 384 bits it can be broken
10:54 easily today number two commercial 512 bits it is breakable by three letter
11:05 organization number three military 1024 bits it is not breakable by anyone on
11:14 this Earth number four Ellen 2048 bits
11:20 it is not breakable by anyone on other planets also since RSA is only used for two
11:30 small computations everyone should use Ellen strength Keys all the
11:38 time the format of a classic pgp message is as shown in figure numerous other
11:47 formats are also in use the message has three parts containing the idea key the
11:56 signature and the message respectively the key part contains the key
12:04 identifier and key itself the signature part contains a
12:10 signature header followed by a time stamp next field is the identifier for
12:18 the senders public key that can be used to decrypt the signature hash and this
12:26 field is followed by type information that identifies the
12:32 algorithms used and the encrypted md5 hash
12:40 itself the message part also contains a header followed by the default file name
12:48 to be used if the receiver writes the file to the disk next field is message
12:56 creation Tim stamp and finally the message itself this was about working of pgp

13:06 algorithm now let us see a product that is open pgp which is being used nowadays based

13:16 on pgp algorithm open pgp is the most widely

13:22 used Email encryption standard in the world today it is defined by the open

13:30 pgp working group of the internet engineering task force

13:37 ietf standard RFC 480 the open pgp Alliance is a growing

13:46 group of companies and other organizations that are implementers of

13:52 the open pgp proposed standard the alliance works to facilitate technical

14:02 interoperability and marketing Synergy between open pgp implementations the open pgp standard

14:11 was originally derived from pretty good privacy pgp protocol RFC 480 document

14:22 contains all the necessary information to develop interoperable applications

14:29 based on the open pgp format it describes the format and methods needed

14:37 to read check generate and write conforming encrypted messages keys and

14:47 signatures in July 1997 pgp incorporation proposed to the ietf that

14:56 there should be a standard called open pgp they gave the ietf permission to use

15:04 the name open pgp to describe this new standard as well as any program that

15:12 supported the standard the ietf accepted the proposal

15:18 and started the open pgp working group open pgp is on the internet

15:26 standards track and is under active development specification of pgp was RFC

15:35 1991 but the current specification in open pgp is RFC

15:42 480 in 2007 the successor of RFC

15:49 24 the standard was extended to support chameleon Cipher by RFC 581 in 2009 and

16:00 encryption based on elliptic curve cryptography as RFC 637 in

16:11 2012 open pgps encryption can ensure secure delivery of files and messages as

16:20 well as provide verification of who created or sent the message using a

16:27 process called digital signing using open pgp for communication

16:35 requires participation by both the sender and the

16:41 recipient open pgp can also be used to secure sensitive files when they are

16:48 stored in vulnerable places like mobile devices or in the

16:56 cloud so after pgp let us move on to privacy enhanced mail

17:04 pem in contrast to pgp which was initially a oneman show our second email

17:12 security protocol privacy enhanced mail p is an official internet standard and

17:21 described in four rfc's RFC 1421 through rfc4

17:30 424 it is a 1993 ietf proposal for securing email using public key

17:40 cryptography very roughly PM covers the same territory as pgp that is privacy and

17:49 authentication for RFC 822 based email systems nevertheless it also has some

17:58 differences from pgp in approach and Technology PM is the outgrowth of work
18:06 by the privacy and security research group psrg of the internet research task force
18:17 irtf messages sent using p are first converted to a canonical form so that
18:26 they all have the same conventions about white space like tabs and
18:34 spaces next a message hash is computed using md2 or
18:41 md5 then the concatenation of the hash and the message is encrypted using d s
18:52 in light of the known weakness of a 56-bit key this choice is certainly
18:59 suspect the encrypted message can then be encoded with base 64 coding and
19:08 transmitted to the recipient as in pgp each message is
19:15 encrypted with a onetime key that is enclosed along with the
19:22 message the key can be protected either with RSA or with tri triple Dees using
19:33 Ed in pem Key Management is more structured as compared to pgp keys are
19:42 certified by x.509 certificates issued by
19:48 CA which are arranged in a rigid hierarchy starting at a single route the
19:56 advantage of this scheme is that certific ific at revocation is possible by having the
20:05 periodically the only problem with p is that nobody ever used it and it has gone
20:13 to that big bit bin in the sky although PM became an ietf proposed standard
20:24 still it was never widely deployed or used the problem was largely
20:32 political who would operate the route and under what conditions there was no
20:38 shortage of candidates but many people were afraid to trust anyone company with
20:47 the security of the whole system the most serious candidate RSA security
20:55 incorporation wanted to charge per certificate issued however some organizations
21:04 balked
21:12 at this idea in particular the US government is allowed to use all us
21:20 patents for free and companies outside the US had become accustomed to use the
21:29 RSA algorithm for free as the company forget to patent it outside the the
21:37 US no one was ready about suddenly having to pay RSS security
21:45 incorporation for doing something that they had always done for free in the end
21:52 no route could be found and PM collapsed in addition to being an
22:00 obstacle to deployment the single rooted hierarchy was rejected Ed by some
22:06 commentators as an unacceptable imposition of central
22:14 Authority this led Phil Zimmerman to propose the web of trust as the pki
22:20 infrastructure for the encryption program pretty good privacy
22:28 pgp efforts to deploy PM were finally abandoned in response to the need to to
22:35 extend the protocol to support mime this led to the development of the
protocol mime object Security Services moss and S mime which shares defao

22:44 standard status with pgp this is an ietf standard a result of
22:51 a group working for a long time the basic idea is to have privacy by virtue
22:59 of hierarchial authentication a receiver trusts the message of the sender when it is
23:06 accompanied by a certificate from his trusted Authority these authoritative
23:14 certificates are distributed from a group called internet policy
23:20 registration Authority IP and policy certificate Authority PCA
23:29 these trusted Authority actually certifies the public key sent by senders
23:37 now after PM let us see third email security algorithm that is secure
23:45 multi-purpose internet mail extensions s mime ietf next venture into email
23:54 security known as secure multi-purpose internet mail extension s mime is
24:02 described in RFC 2632 through
24:09 2643 as we know internet email messages consist of two parts the header and the
24:17 body the header forms a collection of field or value pairs structured to
24:25 provide information essential for the transmission of the message the structure of
these headers can be found
24:34 in RFC 822 the body is normally unstructured
24:41 unless the email is in mind format mime defines how the body of an email message
24:50 is structured the MIM format permits email
24:56 to include enhance Ed text Graphics audio and more in a standardized manner
25:05 via MIM compliant mail systems however mime itself does not provide any
25:13 Security Services the purpose of s mime is to Define such
25:20 services like PM it provides authentication data Integrity secrecy
25:27 and known reputation it also is quite flexible supporting a variety of cryptographic
25:37 algorithms not surprisingly given the name s mime integrates with mime all
25:44 kind of messages to be protected secure multi-purpose internet
25:51 mail extensions is a protocol that adds digital signatures and an encryption to
25:59 internet mime messages a variety of new mime headers are defined for example for
holding
26:09 digital signatures ietf definitely learn something from the PM
26:18 experience s mime does not have a rigid certificate hierarchy beginning at a
26:25 single root instead user can have multiple trust anchors as long as
26:33 certificate can be traced back to some trust anchor the user believes in it is
26:41 considered valid SM mime uses the standard algorithms and
26:50 protocols now in last let us see comparison of email security protocols
26:58 we have studied three email security protocols first was pgp that is pretty
27:06 good privacy second was privacy enhanced mail pem and third was s
27:16 mime as already discussed P could not gain popularity due to its shortcomings

27:24 so it got collapsed pgp and S MIM which was adopted as an internet standard are
27:33 two official email security systems which are currently National Institute
27:40 of Standards and Technology nist specified standards both are considered to be
27:47 secure mail transmission techniques there has been a great deal of confusion in
public and even within
27:57 the IETF about the status of S M and
28:03 pgp let us compare s MIM version 3 and open pgp protocol which are widely used
28:13 nowadays please see the figure although they offer similar services to the users
28:20 the two protocols have many differences s mime supports binary
28:26 format whereas as open pgp supports pkcs7 mime format further and more
28:36 important to corporate users they have different formats for their
28:42 certificates as you can see s mime supports binary certificate based on
28:49 x.509 version 3 whereas open pgp supports previous pgp based bind
28:59 certificates this means that user of one protocol cannot communicate with the
29:05 user of other and also they cannot share authentication
29:13 certificates s mime supports Dey Helman with DSS or RSA but open pgp supports
29:22 elg1 with DSS signature algorithm both both the protocol supports S1 hash
29:32 algorithm s mime supports CMS format of mime encapsulation of signed data but
29:41 open pgp supports asy the difference between the two protocols is similar to the
differences
29:51 between gif and JPEG file they both do basically the same thing for end users
29:59 but their formats are very different so friends this was about protocols for
30:08 email Security in this lecture after covering brief introduction about email
30:15 security we discussed different email security protocols like pretty good
30:23 privacy pgp privacy enhanced mail p and secure multi-purpose internet mail
30:33 extension s mime in last we also compared to most commonly used protocols
30:41 s MIM version 3 and open pgp thank you
30:50 [Music]
31:11 [Music]